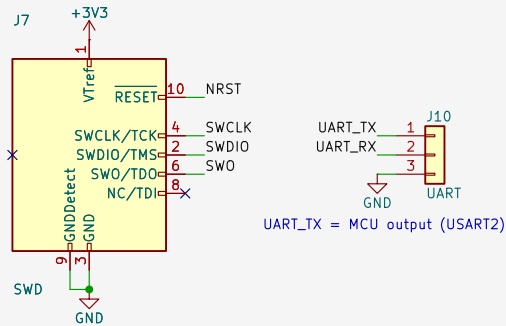
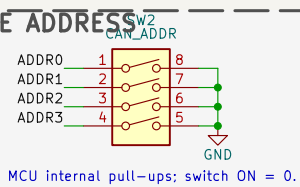


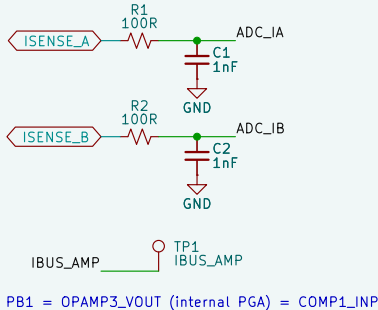
DEBUG: SWD + UART



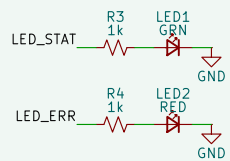
CAN NODE ADDRESS



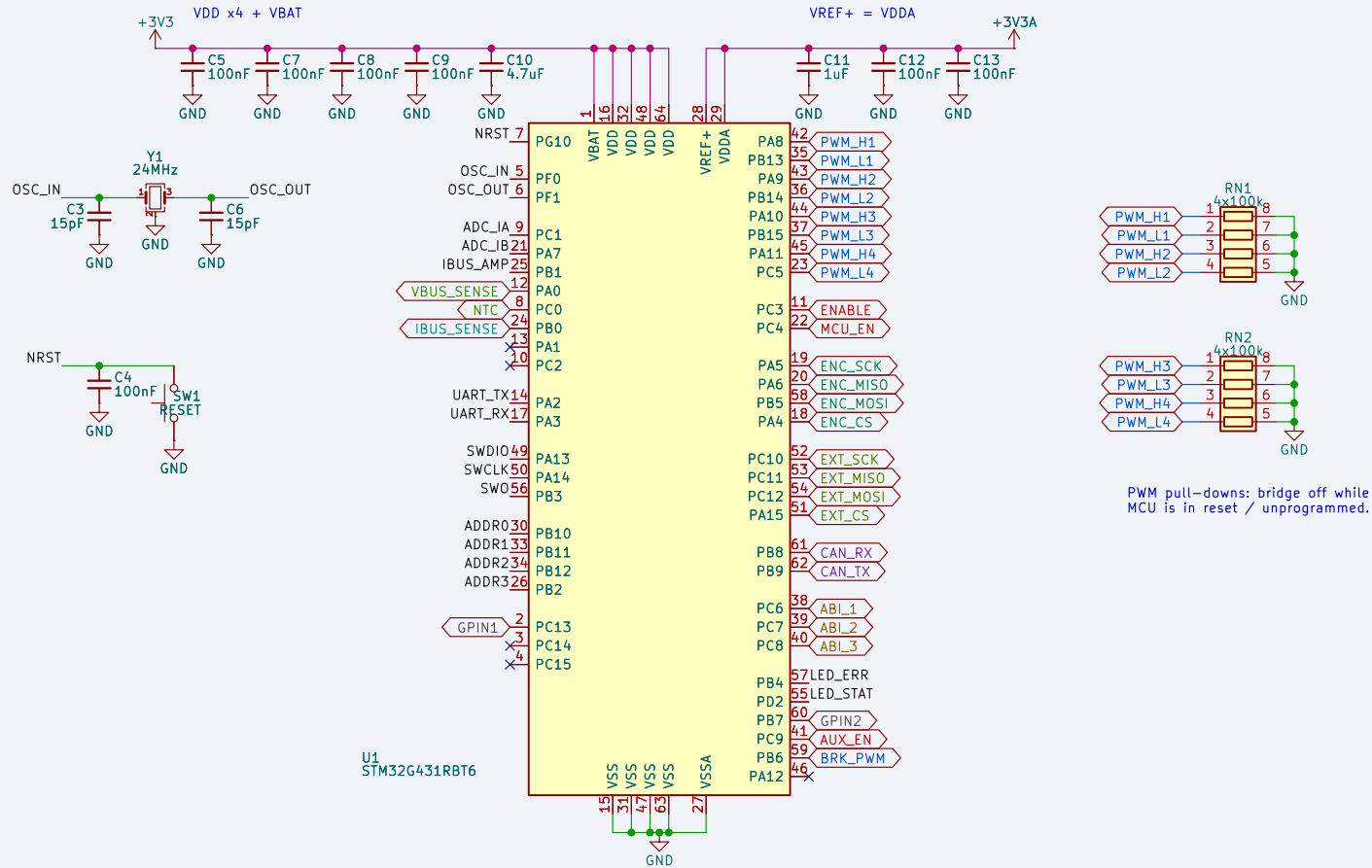
ADC INPUT RC



STATUS LEDS



MCU STM32G431RBT6



DESIGN NOTES (Rev B)

- Option bytes: nSWBOOT0 = 0 (PB8 = BOOT0 is shared with FDCAN1_RX).
- TIM1 complementary PWM, CH1-CH4 + CH1N-CH4N, centre aligned 30 kHz (20-50 kHz). DTG 50-100 ns is a backup only; DRV8300D inserts a fixed 200 ns dead time.
- Duty limit 97-98 % (bootstrap refresh). Pre-charge bootstraps (low sides on) after every enable and after Hi-Z.
- Hardware trips: RS3 -> PB0 OPAMP3 (PGA) -> PB1 -> COMP1 -> TIM1_BKIN. IA PC1 -> COMP3, IB PA7 -> COMP2 (DAC thresholds). ENABLE PC3 -> TIM1_BKIN2.
- ADC: IA PC1 = ADC1_IN7 and IB PA7 = ADC2_IN4 (simultaneous). VBUS PA0, NTC PC0.
- No brake chopper fitted: limit regen via VBUS monitoring (Q9/J9 footprint is DNP).
- External encoder uses SPI3 (SPI2 SCK only on PB13/PF1, both in use).
- ABI / Hall on TIM3 CH1-CH3 (encoder mode or Hall XOR interface).

Power Supply

File: power_supply.kicad_sch

Power Stage

File: power_stage.kicad_sch

Interfaces

File: interfaces.kicad_sch

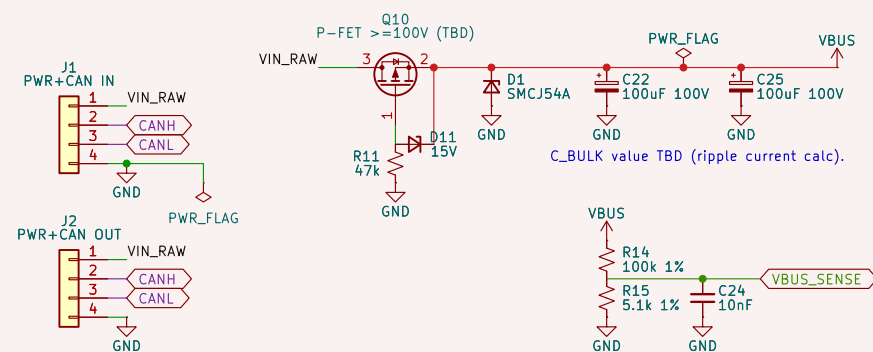
FOC Motor Driver

Sheet: /
File: foc-driver.kicad_sch

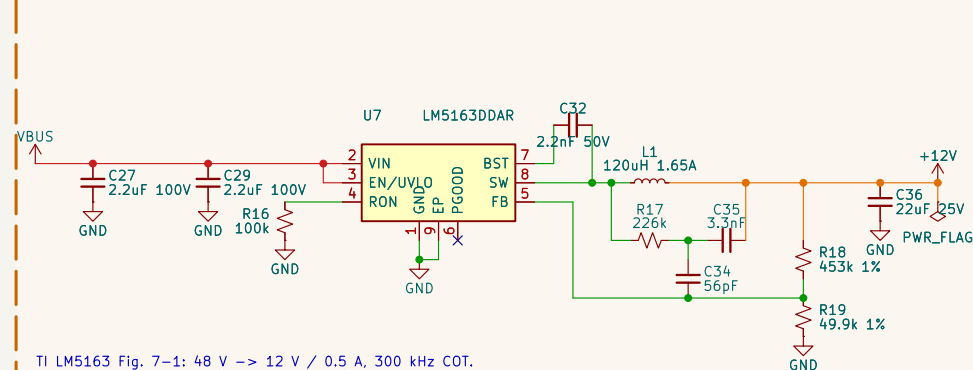
Title: FOC Motor Driver Rev B - MCU

Size: A3 Date: 2026-10-02
KiCad E.D.A. 10.0.6

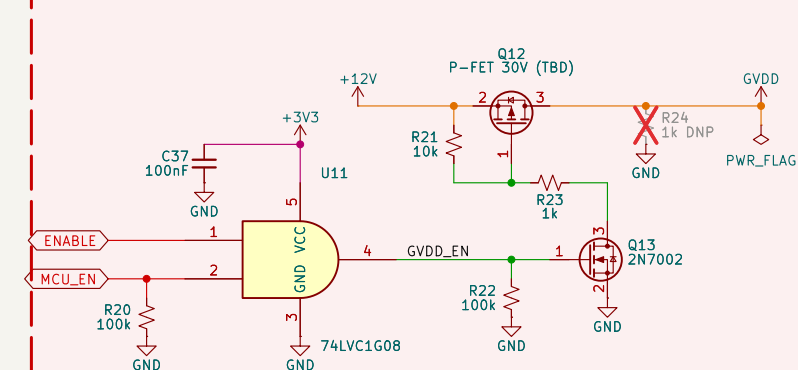
Rev: B
Id: 1/4



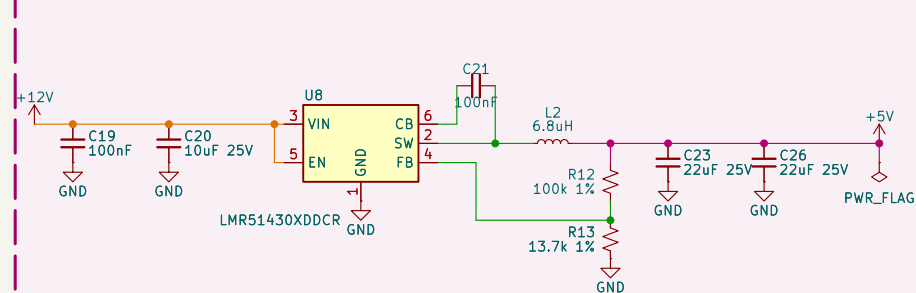
Daisy chain: J1/J2 in parallel.
Current rating = whole chain.
Pinout proposal (connector TBD).



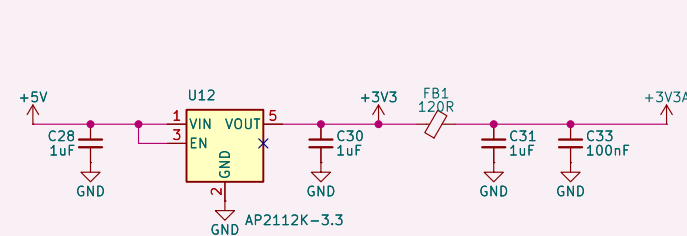
TI LM5163 Fig. 7-1: 48 V $\rightarrow$ 12 V / 0.5 A, 300 kHz COT.
VIN/EN abs max 100 V (TVS clamp ~87 V).



ENABLE low OR MCU_EN low -> GVDD off -> DRV8300 UVLO -> all FETs off
(independent of MCU; TIM1_BKIN2 is the fast path).
GVDD decay time set by driver caps; R DNP = optional bleeder.

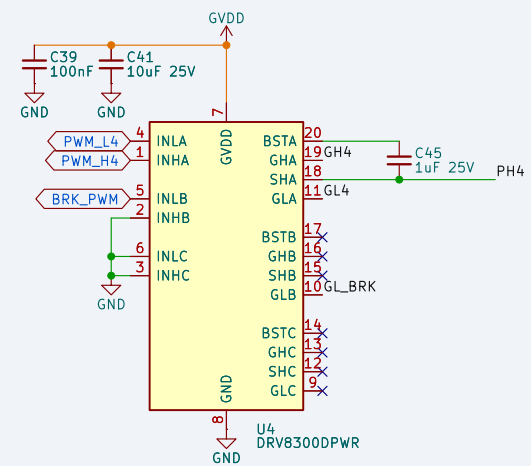
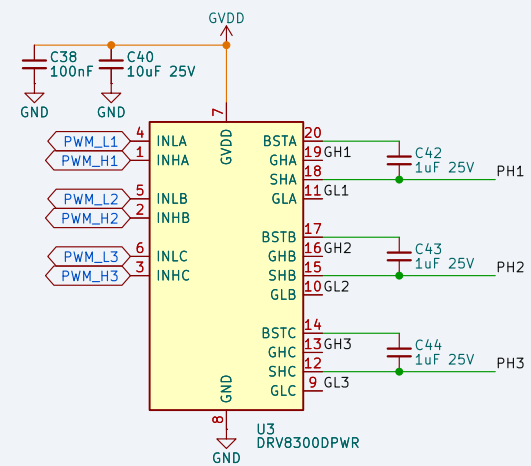


500 kHz variant, datasheet Table 9–1 (5 V: 6.8 μ H, 2x22 μ F, 100k/13.7k).
5 V feeds CAN transceiver VCC and J8 encoder supply.



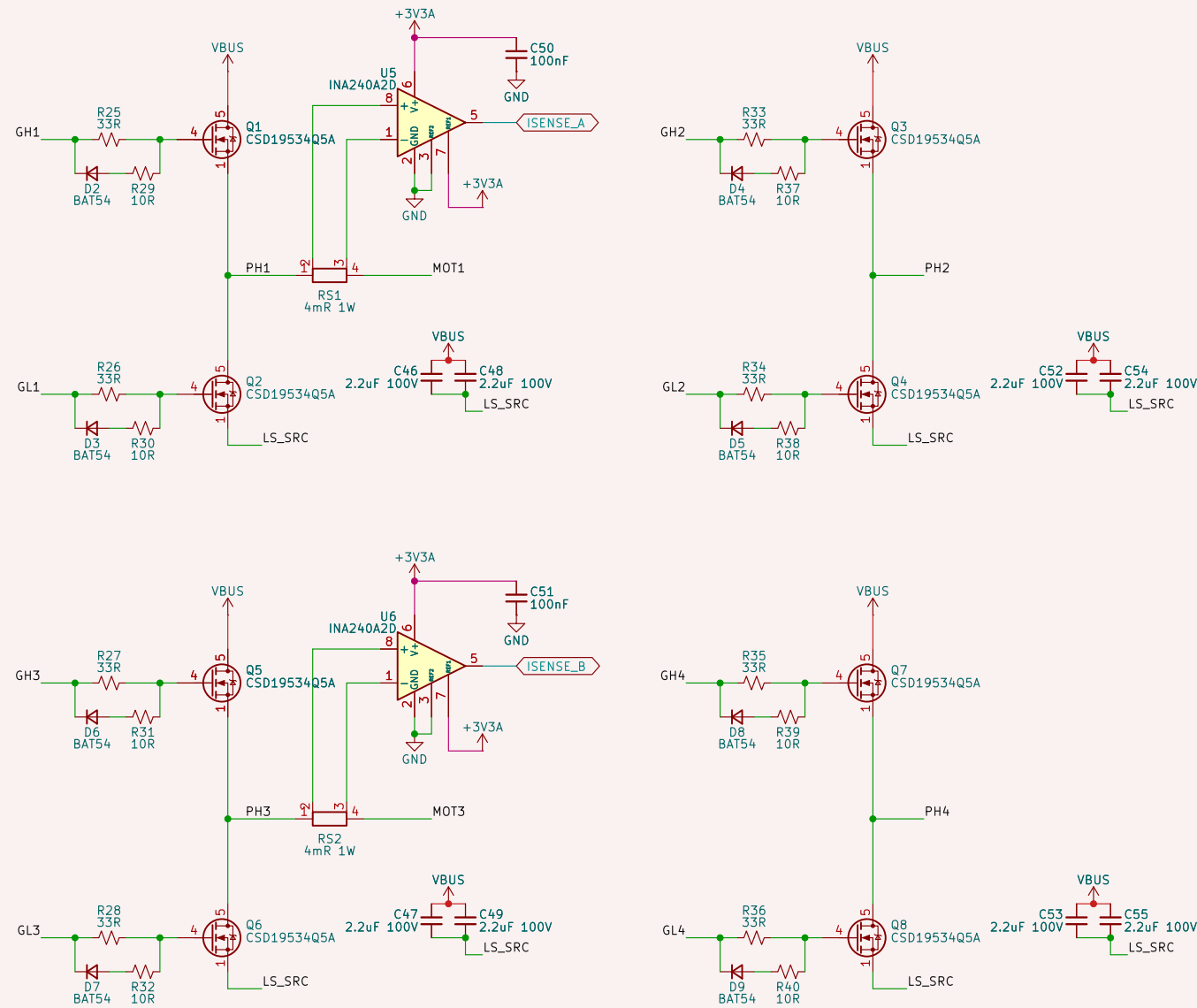
- +3V3: MCU, CAN VIO, logic (~80 mA).
- +3V3A: VDDA/VREF+, INA240, AS5047D, NTC.

GATE DRIVERS

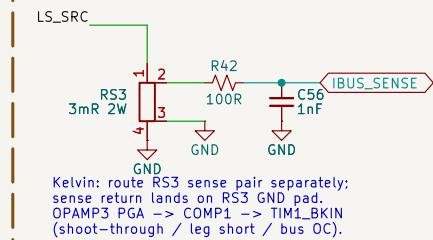


U3 = HB1+HB2+HB3 (BLDC U/V/W on one driver).
U4 = HB4 (+ optional brake chopper on ch. B).
BLDC build: U4, Q7, Q8 + HB4 parts DNP.
DRV8300D TSSOP: fixed 200 ns dead time + lockout.

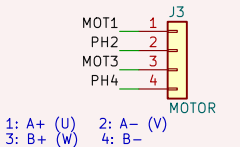
HALF BRIDGES + IN-LINE CURRENT SENSE



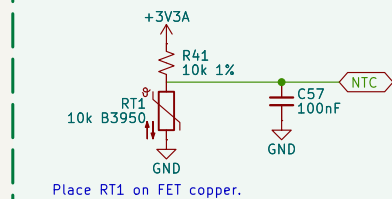
BUS SHUNT RS3



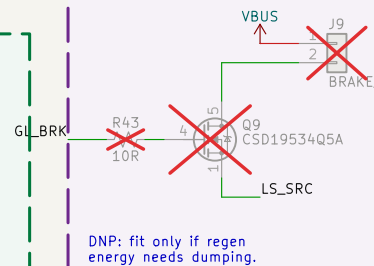
MOTOR



FET TEMP



BRAKE CHOPPER (DNP)



FOC Motor Driver

Sheet: /Power Stage/
File: power_stage.kicad_sch

Title: Power Stage

Size: A3 Date: 2026-10-02
KiCad E.D.A. 10.0.6

Rev: B
Id: 3/4

